

Maths

Unit 4 - Addition and
Subtraction

WALT:

6Nc.02 - Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.



**With support, I
can answer
questions that use
letters to
represent
quantities.**



**Independently, I
can answer
questions that use
letters to
represent
quantities.**



**I can specialise by
showing all the
possible values of
a variable.**

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

Retrieval Practice

Last Lesson Mr. Rhys says: "6 - y is the same as y - 6". Is he correct? Explain your answer.	Last Week Ms. Colette has 500k VND in her bank account. She buys a bracelet for 875k VND. How much money does she have in her bank account?
Last Unit Ms. Alice created a sequence: 4, 8, 12, 16, 20, 24 She says, "The position to term rule is +4." Is she correct?	Last Year 8 What are the possible values of  and  when  +  +  +  = 1.3 kg?

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

Pg. 58 - 61

- 1 Cheng plays a board game using a dice.

He uses the instructions together with his dice score to work out how many spaces he moves.

d represents the dice score.

For example:

Score	Calculate	Spaces moved
		

Work out how many

	Score
a	
b	
c	

- 4 Martha buys 2 more pairs of socks than shoes.



- a Copy and complete the table where x represents the number of pairs of shoes and y represents the number of pairs of socks.

x	1	2	
y	3		6

- b Write a number sentence linking x , y and 2.

- 5 This puzzle has 9 pieces.



Hassan places 1 or more shapes and Sanjay places the other shapes.

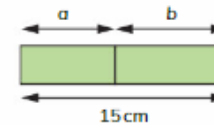
- a Copy and complete this table to show the number of pieces each person places.

x (number of pieces placed by Hassan)									
y (number of pieces placed by Sanjay)									

- 6 a and b represent the lengths, in centimetres, of two strips of card. b is 3 cm longer than a .

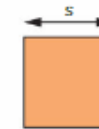
The two strips are placed end to end.

The total length is 15 cm.



Find the values of a and b .

- 7 The perimeter (p) of a square is the sum of the lengths of the sides. s represents the length, in centimetres, of a side.



$$p = s + s + s + s$$

- a What is the value of p when $s = 5$?
 b What is the value of p when $s = 7$?
 c What is the value of s when $p = 32$?
 8 This isosceles triangle has a perimeter of 15 cm.

x and y represent the lengths, in centimetres, of the sides of the triangle shown in the diagram.

Think like a mathematician

a , b and c each represent a whole number from 1 upwards.

$$a + b + c = 7$$

Find all the possible values for a , b and c .
 How many different solutions can you find?

You will show you are specialising when you find all the possible values for a , b and c .

- a Find three possible sets of a , b and c .
 b Write a formula for the perimeter of the triangle.

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

- 1 Cheng plays a board game using a dice.

He uses the instructions together with his dice score to work out how many spaces he moves.

d represents the dice score.

For example:

Score	Calculate	Spaces moved
	$d + 4$	9 spaces

Work out how many spaces Cheng moves.

	Score	Calculate	Spaces moved
a		$d - 3$	3
b		$6 - d$	4
c		$4 + d$	7

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

2 For each pair of expressions write 'equal' or 'not equal'.

a $d + 4$ $4 + d$ **equal**

b $2 + d$ $d + 2$ **equal**

c $5 - d$ $d - 5$ **not equal**

3 Khalid says, ' $d + 3$ is the same as $3 + d$ so $d - 3$ must be the same as $3 - d$.' Is Khalid correct? Explain your answer.

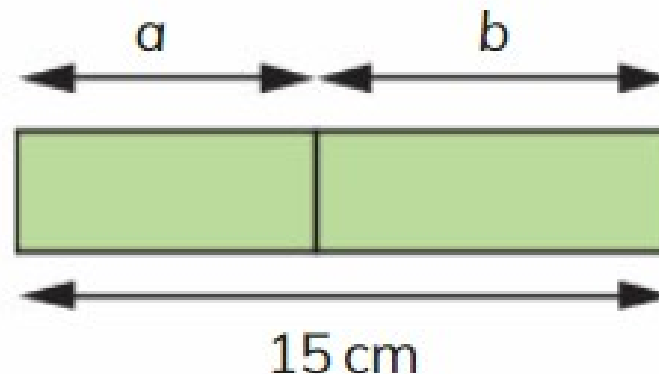
Khalid is incorrect. Due to the commutative law it does not matter which number is written first in addition questions but it does in subtraction questions.

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

- 6 a and b represent the lengths, in centimetres, of two strips of card.
 b is 3 cm longer than a .

The two strips are placed end to end.

The total length is 15 cm.



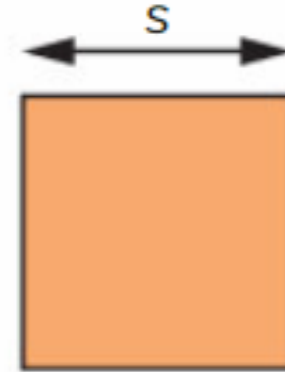
$$\begin{aligned} a &= 6\text{cm} \\ b &= 9\text{cm} \end{aligned}$$

Find the values of a and b .

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

7 The perimeter (p) of a square is the sum of the lengths of the sides.

s represents the length, in centimetres, of a side.



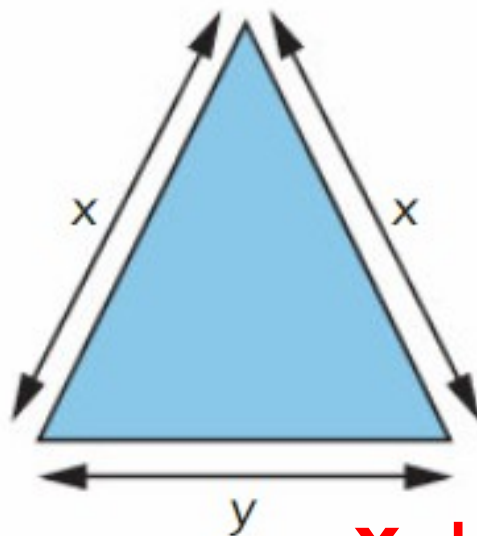
$$p = s + s + s + s$$

- a** What is the value of p when $s = 5$? **20**
- b** What is the value of p when $s = 7$? **28**
- c** What is the value of s when $p = 32$? **8**

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

8 This isosceles triangle has a perimeter of 15 cm.

x and y represent the lengths, in centimetres, of the sides of the triangle shown in the diagram.



$$x + x + y = 15$$

- a Find three possible sets of values for x and y .
- b Write a formula for the perimeter (p) of the triangle using x and y .

$$Y = 13$$

$$X = 1$$

$$Y = 11$$

$$X = 2$$

$$Y = 9$$

$$X = 3$$

$$Y = 7$$

$$X = 4$$

$$Y = 3$$

$$X = 6$$

$$Y = 1$$

$$X = 7$$

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

Homework

Maths - Workbook - Pages 50 - 54

ESL - Workbook pages 20 - 23

Due - Thursday 24th September

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

Think like a mathematician

a , b and c each represent a whole number from 1 upwards.

$$a + b + c = 7$$

Find all the possible values for a , b and c .

How many different solutions can you find?

You will show you are specialising when you find all the possible values for a , b and c .

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.



**With support, I
can answer
questions that use
letters to
represent
quantities.**



**Independently, I
can answer
questions that use
letters to
represent
quantities.**



**I can specialise by
showing all the
possible values of
a variable.**

WALT: Recognise the use of letters to represent quantities that vary in addition and subtraction calculations.

Plenary

If $5b = 20$ what does $b =$?

$b = 4$

Why do you think we don't use the multiplication sign "x" ?